# IRP "Data-driven simulation of soccer players and teams"

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## Abstract

Learning realistic multi-agent behavior from offline data is a challenging problem due to limited interaction, partial observability, and complex agent coordination. In this work, we study the problem of learning multi-agent policies from spatiotemporal sports data using offline imitation learning and reinforcement learning approaches. We focus on evaluating the ability of different methods to reconstruct realistic trajectories and generalize to unseen states without environment interaction. To this end, we propose an evaluation framework based on trajectory reconstruction and behavioral similarity. Our analysis highlights the limitations of existing approaches in capturing coordinated behavior under partial observability and emphasizes the importance of incorporating domain-specific constraints for achieving realistic and interpretable multi-agent policies.

## 1 Introduction

In recent years, the availability of spatio-temporal tracking and event data has enabled the development of data-driven approaches for modeling player and team behavior in soccer. In particular, imitation learning has emerged as a powerful paradigm for learning policies directly from expert demonstrations without requiring explicit reward design (Foster et al., 2024). At the same time, reinforcement learning and its multi-agent extension (MARL) provide a general framework for sequential decision-making in environments involving multiple interacting agents (Zhang et al., 2021).

Modeling behavior in team sports naturally leads to a multi-agent setting, where agents must act under partial observability and coordinate their actions over time. Such problems are commonly formulated as decentralized partially observable Markov decision processes (Dec-POMDPs), which capture the challenges of limited information and decentralized decision-making (Fujii et al., 2020). Recent advances in centralized training with decentralized execution (CTDE) further highlight the importance of leveraging global information during training while maintaining decentralized policies at execution time (Amato, 2024).

Despite these advances, it is still unclear how effectively offline learning methods can model both individual and coordinated behavior from logged data, particularly in complex real-world domains such as sports. Existing work often focuses on separate aspects, such as trajectory imitation or reward-based optimization, without evaluating them in a consistent and unified way.

This work aims to address this gap by investigating the following research questions:

- **Imitation fidelity:** To what extent can a policy reproduce player behavior from logged data?
- **Tactical realism:** To what extent can a policy capture coordinated multi-agent behavior?
- **Generalization and robustness:** To what extent does the learned policy remain plausible under new opponents, states, or match contexts?

While individual studies propose methods addressing specific aspects of these questions, this work adopts a comparative perspective, analyzing different learning paradigms and their respective strengths and limitations. In particular, the goal is to provide a structured evaluation of existing methods and to examine how their performance can be assessed consistently across imitation fidelity, tactical realism, and generalization.

## 2 Background

The following section provides the necessary theoretical context for this analysis. We review key research areas relevant to the problem setting, including imitation learning, offline reinforcement learning, multi-agent reinforcement learning, and data-driven modeling of multi-agent behavior in sports.

### 2.1 Imitation Learning

Imitation learning (IL) provides a natural framework for learning decision-making policies directly from expert demonstrations without requiring explicit reward signals. Instead of relying on interaction with an environment, the learner aims to mimic expert behavior based on observed trajectories, making IL particularly suitable for offline settings where only pre-collected data is available (Foster et al., 2024).

In many real-world domains, including sports analytics, large-scale datasets provide detailed spatio-temporal information such as player positions and ball trajectories. This enables the construction of policies that replicate the behavior of expert agents, allowing the learning of coordinated multi-agent strategies directly from real-world data. In this context, imitation learning offers a practical approach to modeling complex interactions by leveraging demonstrations of professional players.

The most widely used approach to imitation learning is behavioral cloning, which formulates the problem as supervised learning by predicting expert actions given observed states (Foster et al., 2024). Its simplicity and compatibility with modern function approximation methods, such as deep neural networks, make it highly effective in practice. However, a key limitation of behavioral cloning is its sensitivity to distributional shift, where small deviations from the expert policy can lead to compounding errors during rollout (Foster et al., 2024).

Despite these challenges, imitation learning remains a strong baseline for offline policy learning, as it avoids the need for environment interaction and allows direct utilization of real-world datasets. This makes it particularly suitable for settings where learning must rely entirely on offline data.

### 2.2 Offline Reinforcement Learning

Offline reinforcement learning (offline RL) extends the reinforcement learning paradigm to settings where the agent learns exclusively from a fixed dataset, without interacting with the environment during training. Similar to imitation learning, offline RL relies on real-world data, such as recorded trajectories of agents, including states and actions. However, a key difference is that reinforcement learning additionally leverages reward signals to guide policy optimization. Formally, the offline RL setting assumes access to a dataset of trajectories generated by a behavior policy, and the objective is to learn a target policy that maximizes the expected return.

As described in (Kumar et al., 2020), the goal is to learn a policy $\pi^*$ that maximizes the expected cumulative reward. In contrast to imitation learning, which focuses on reproducing observed actions, offline RL aims to go beyond the behavior in the dataset by learning which actions lead to better long-term outcomes. However, applying offline RL in domains such as football is not straightforward. One of the main challenges is that rewards are often sparse and delayed.

In many cases, meaningful feedback is only available at the end of a sequence (e.g., whether a team scores or wins), making it difficult to understand the impact of individual actions during the game. In addition, defining a suitable reward function in football is inherently difficult.

While final outcomes such as goals provide clear signals, many important actions—such as passes, positioning, or defensive movements—contribute indirectly to success and are harder to quantify. As a result, practical approaches often rely on reward shaping, where intermediate signals (e.g., successful passes or ball progression) are used to provide additional guidance. Despite these challenges, offline RL remains a promising approach for learning from real-world sports data. By combining trajectory information with appropriately designed reward signals, it becomes possible to learn policies that not only mimic observed behavior but also capture longer-term strategic aspects of the game.

### 2.3 Multi-Agent Reinforcement Learning in Sports

Multi-agent reinforcement learning (MARL) considers settings where multiple agents interact

within a shared environment and learn their policies simultaneously. In contrast to single-agent settings, each agent's behavior depends not only on the environment but also on the actions of other agents, which creates strong interdependencies and makes learning more challenging (Zhang et al., 2021).

Sports analytics provides a natural application domain for MARL, as it involves multiple interacting agents whose actions are tightly coupled over time. In football, this corresponds to 22 players acting in parallel, where the behavior of each player depends on both teammates and opponents.

Recent advances in tracking technologies have enabled the collection of large-scale spatiotemporal datasets, allowing coordinated behavior to be learned directly from demonstrations (Le et al., 2017). However, coordination in such data is not explicitly labeled and must be inferred from observed trajectories.

In addition, player behavior is often structured around roles (e.g., defenders or midfielders), which are dynamic and may change throughout a game. Another important challenge is partial observability. In realistic settings, each agent has access only to local information, making it necessary to learn decentralized policies while still achieving coordinated team behavior. Most existing approaches focus on trajectory prediction or short-term behavior modeling, rather than learning underlying decision-making policies.

As a result, they often capture local movement patterns but fail to represent coordinated strategies or realistic multi-agent interactions. These observations motivate the need for a systematic comparison of different learning approaches and their ability to model both individual and coordinated behavior in multi-agent settings.

## 3 Related Work

In this section, we review the main methodological approaches considered in this work. The focus is on methods that can learn from offline data and are relevant for modeling player and team behavior in soccer. The literature reviewed here was identified using academic search tools including Google Scholar, Consensus, Perplexity, and Elicit. For each method, we discuss its main idea, strengths, and limitations, and explain its relevance for the objectives of this thesis.

In addition to search tools, particular attention was given to publications from top-tier machine learning and reinforcement learning venues, including NeurIPS, ICML, and ICLR, which are widely recognized for high-impact research in these areas. For domain-specific applications in sports analytics, relevant work from specialized venues such as the MIT Sloan Sports Analytics Conference and related workshops was also considered.

These venues were selected because they represent the most influential and relevant research communities for this topic, ensuring that the reviewed methods reflect both state-of-the-art algorithmic developments and practical applications in sports settings. Priority was given to highly cited and widely used approaches, as well as recent work that addresses challenges in offline learning and multi-agent systems.

### 3.1 Imitation Learning Methods

Imitation learning provides a natural starting point for learning from recorded data, as it directly models expert behavior without requiring reward signals (Foster et al., 2024; Tiapkin et al., 2024). The most common approach is behavioral cloning (BC), which formulates the problem as supervised learning over state-action pairs (Foster et al., 2024).

Behavioral cloning is simple and stable to implement and often performs well when the test data is similar to the training distribution. However, it suffers from compounding errors due to distributional shift, where small prediction mistakes accumulate over time and lead to unrealistic trajectories (Ho and Ermon, 2016; Tiapkin et al., 2024). This limitation makes it difficult to capture long-term dependencies and coordinated behavior.

To address these issues, more advanced methods such as Generative Adversarial Imitation Learning (GAIL) have been proposed (Ho and Ermon, 2016). GAIL learns a policy by matching the distribution of expert trajectories using an adversarial objective, leading to more realistic and consistent behavior. It can be applied in both online and offline settings(Liu et al., 2023). However, GAIL introduces additional complexity, requires careful training, and often depends on interaction with an environment or additional datasets. Moreover, scaling GAIL to multi-agent coordination remains challenging (Yang et al., 2020).

Overall, imitation learning methods are effective for reproducing individual behavior (imitation

fidelity), but are limited in capturing long-term strategy and generalization.

### 3.2 Offline Reinforcement Learning Methods

Offline reinforcement learning extends imitation learning by incorporating reward signals and optimizing long-term outcomes (Tiapkin et al., 2024; Zhuang et al., 2023). Instead of only mimicking observed actions, these methods aim to learn which actions lead to better results over time.

A key challenge in offline RL is distributional shift, where the learned policy encounters state-action pairs that are not well covered in the dataset (Zhuang et al., 2023). Conservative Q-Learning (CQL) addresses this issue by penalizing out-of-distribution actions and preventing overestimation (Zeng, 2025) . This makes CQL robust to noisy or imperfect datasets, but also leads to conservative behavior and sensitivity to hyperparameters.

Another recent direction is sequence modeling approaches such as the Decision Transformer (Chen et al., 2021). Instead of learning value functions or policies explicitly, it models trajectories as sequences and predicts actions conditioned on desired returns . This allows it to capture long-term dependencies and handle sparse rewards more effectively. However, it requires large datasets and careful tuning of sequence length and conditioning.

In this work, when reward-based methods are used, we adopt the reward structure from the Google Research Football environment (Kurach et al., 2020). This includes sparse rewards (e.g., scoring goals) and can be extended with reward shaping to incorporate intermediate signals such as ball progression or successful actions. This choice provides a consistent and realistic evaluation setting for all reward-based methods.

Overall, offline RL methods are better suited for capturing long-term decision-making and generalization, but depend heavily on reward design and dataset quality.

### 3.3 Multi-Agent and Domain-Specific Approaches

Modeling team sports requires capturing interactions between multiple agents. While general RL methods can be applied, domain-specific approaches provide additional insights (Zhang et al., 2021) .

One such approach is data-driven ghosting, which reconstructs team behavior by comparing actual trajectories to expected or "average" behavior . These methods are highly interpretable and useful for tactical analysis, as they capture team-level structure and coordination. However, they are not true learning methods and cannot generate new behaviors beyond the observed data.

In contrast, reinforcement learning methods such as Proximal Policy Optimization (PPO) can learn coordinated multi-agent policies. PPO has been shown to achieve strong performance in cooperative multi-agent environments, including Google Research Football (Yu et al., 2022) . It provides stable policy updates and good performance in sequential decision-making tasks. However, PPO requires interaction with the environment and is not naturally suited for purely offline settings, making its application in this work limited to adapted or evaluation scenarios. Recent work also shows that PPO-style methods can be adapted to offline settings under certain constraints (Zhuang et al., 2023) .

These methods highlight an important trade-off: domain-specific approaches offer realism and interpretability, while learning-based methods provide flexibility and generalization.

### 3.4 Reward-Based Imitation Learning (Contrast Class)

Inverse imitation learning methods aim to infer the underlying reward function that explains expert behavior (Arora and Doshi, 2020). This provides a principled framework for understanding decision-making and can improve generalization.

However, inverse reinforcement learning is often computationally expensive, sensitive to modeling assumptions, and difficult to scale to complex multi-agent environments (Arora and Doshi, 2020) . In addition, the inferred reward function is not always unique, which makes interpretation challenging.

For these reasons, inverse imitation learning is not considered as a primary method in this work, but rather as a conceptual reference point. It provides useful insight into the role of reward modeling, but is less practical for large-scale offline multi-agent settings.

### 3.5 Summary

Based on these observations, the methods considered in this work are selected to cover different aspects of the research questions. Behavioral cloning is included as a baseline for imita-

tion fidelity, while GAIL extends this approach by improving trajectory-level realism. Data-driven ghosting and PPO are considered for modeling coordinated multi-agent behavior and capturing team-level structure. Finally, Decision Transformers and offline reinforcement learning methods such as CQL are used to study generalization and robustness under unseen conditions.

Overall, these methods are not selected to identify a single best approach, but rather to represent different modeling paradigms and enable a structured comparison across imitation fidelity, tactical realism, and generalization.

This motivates the need for a unified evaluation framework, which is the main focus of this work. In particular, we distinguish between evaluation under imitation settings with available ground truth and evaluation under generalization scenarios without ground truth.

## 4 Evaluation Protocol

This section describes the evaluation framework used in this work to assess the effectiveness of the considered methods. The goal is to provide a structured and consistent way to evaluate whether the learned policies successfully address the research questions defined earlier. In particular, we aim to evaluate how well different methods reproduce individual player behavior, capture coordinated multi-agent interactions, and generalize to previously unseen situations.

Following the discussion in the previous section, we distinguish between two evaluation scenarios. The first corresponds to an imitation setting, where ground truth data in the form of expert trajectories is available and can be used for direct comparison. The second corresponds to a generalization setting, where policies are evaluated in unseen or simulated scenarios for which no ground truth is available. In this case, evaluation relies on indirect or proxy measures of performance and realism.

To ensure a consistent comparison across methods, the evaluation framework is organized into four main components: datasets, evaluation scenarios, metrics, and experimental setup.

### 4.1 Datasets

A key observation guiding this work is that existing sports datasets often provide either rich action and contextual information or detailed spatio-temporal tracking data, but rarely both simultaneously. This creates a challenge for learning realistic policies, as a complete representation of the environment requires both state information and action descriptions.

To address this limitation, we adopt a multi-dataset approach, combining different data sources to construct a comprehensive representation of player behavior.

**Dataset A (DFL).** The primary dataset used in this work is an integrated dataset of spatio-temporal tracking and event data from professional soccer matches (Bassek et al., 2025). It contains synchronized information about player and ball positions as well as annotated events such as passes, shots, and fouls. This makes it particularly suitable for learning and evaluating multi-agent policies.

**SoccerTrack v2.** SoccerTrack v2 provides a structured dataset with explicit representations of state, action, and context, facilitating consistent modeling and evaluation (Scott et al., 2025). Its clean structure simplifies preprocessing and supports robust experimentation.

**Metrica Sports Dataset.** The Metrica Sports dataset provides publicly available tracking data, including player and ball positions over time (Metrica Sports, 2019). It is primarily used for debugging and baseline experiments due to its simplicity and accessibility.

**UIED-style Data.** We also consider datasets based on unified event taxonomies for invasion games, which provide structured annotations of low-level and high-level events (Biermann et al., 2021). These datasets are useful for modeling event semantics but are limited in trajectory modeling.

**BallRadar.** BallRadar is used as a reference approach for trajectory-based modeling, focusing on reconstructing player movements rather than learning decision-making policies (bal, 2023).

**Event-based Datasets (Wyscout, StatsBomb).** Event-based datasets such as the Wyscout dataset (Pappalardo et al., 2019) and StatsBomb Open Data (StatsBomb, 2019) provide large-scale annotated logs of match events, including passes, shots, fouls, and duels. These datasets contain detailed information about the type, timing, location, and outcome of each action, enabling fine-grained analysis of decision-making and tactical behavior.

In particular, the Wyscout dataset covers multiple professional competitions and includes millions of annotated events, making it one of the largest publicly available soccer datasets (Pappalardo et al., 2019). StatsBomb complements this by providing rich contextual annotations, such as pressure, player roles, and tactical context.

While these datasets are highly valuable for modeling player actions and evaluating imitation fidelity, they do not provide continuous tracking of player positions. Therefore, they cannot fully capture the underlying state of the environment and must be combined with tracking datasets.

**Summary.** Overall, the selected datasets complement each other. Tracking datasets are used to define the state space, while event datasets provide action and contextual information. This combination enables a consistent evaluation across imitation fidelity, tactical realism, and generalization.

### 4.2 Evaluation Methodology

In this work, we evaluate the considered methods with respect to the research questions introduced earlier, namely imitation fidelity, tactical realism, and generalization. To this end, we adopt a unified evaluation perspective in which different methods are assessed through their respective metrics, while the interpretation of these metrics is aligned with the underlying research objectives.

The evaluation is based on a set of representative methods, including Behavioral Cloning (BC), Generative Adversarial Imitation Learning (GAIL), Conservative Q-Learning (CQL), data-driven ghosting, Decision Transformers, and Proximal Policy Optimization (PPO). Although these methods rely on heterogeneous learning paradigms and evaluation metrics, they can be consistently mapped to specific aspects of the research questions.

For imitation fidelity, methods such as Behavioral Cloning and GAIL are evaluated based on their ability to reproduce expert behavior. Concretely, we follow a supervised evaluation protocol using a train-validation-test split of the dataset. Given a state extracted from the test set, the learned policy predicts the corresponding action, which is then compared to the expert action recorded in the data. This is measured using action prediction accuracy (Pomerleau, 1989). In addition, trajectory-level evaluation is performed by comparing generated sequences of states and actions to real-world trajectories, capturing longer-term dependencies and behavioral consistency (Ho and Ermon, 2016).

To assess tactical realism, we extend this evaluation to multi-agent and temporal settings. Instead of predicting a single action, the model is provided with partial trajectories, i.e., sequences of past player positions and actions, and is tasked with predicting their continuation. The predicted trajectories are then compared to the ground truth trajectories using spatial and temporal similarity measures, such as positional distance and movement coherence. This allows evaluating whether the learned policies capture coordinated team behavior and realistic spatial structure, in line with prior work on coordinated multi-agent imitation learning and trajectory modeling (Le et al., 2017).

For generalization, we evaluate the ability of the learned policies to perform in scenarios that differ from the training data. This includes testing on out-of-distribution states or against unseen opponents. In this setting, no ground truth trajectories are available for direct comparison. Instead, evaluation is based on performance metrics such as cumulative reward or win rate, which reflect how well the policy adapts to new situations. This approach follows standard evaluation practices in reinforcement learning, where performance is measured through expected return (Schulman et al., 2017; Kumar et al., 2020). Methods such as CQL (Kumar et al., 2020), Decision Transformers (Chen et al., 2021), and PPO (Schulman et al., 2017) are particularly relevant in this context, as they explicitly optimize for long-term performance and robustness.

To ensure a fair and consistent comparison across methods, all models are trained and evaluated using the same dataset splits and state-action representations. In particular, the dataset is divided into training, validation, and test sets, and all methods are trained exclusively on the training portion. Evaluation is performed on the same held-out test set for imitation and trajectory-based metrics. For generalization, all methods are evaluated under comparable out-of-distribution scenarios, such as unseen game states or opponent configurations. This setup ensures that differences in performance can be attributed to the learning methods rather than variations in data or evaluation conditions.

The planned timeline of the experimental work is illustrated in Figure 1.

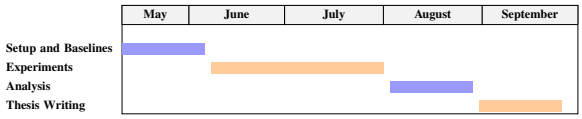



Figure 1: Planned timeline of the experimental and writing phases of the thesis.

Overall, this evaluation framework combines method-specific metrics with consistent experimental procedures, ensuring that each research question is addressed in a quantifiable and reproducible manner, while enabling comparison across different learning paradigms.

## 5 Conclusion

In this work, we investigated the problem of learning realistic and generalizable multi-agent policies from offline demonstration data, with a particular focus on the domain of soccer. By combining imitation learning and offline reinforcement learning approaches, we explored to what extent complex coordinated behaviors can be reconstructed solely from observational data, without direct interaction with the environment.

To address this challenge, we proposed a unified evaluation framework that connects heterogeneous learning paradigms through three key research questions: imitation fidelity, tactical realism, and generalization under distributional shift. This framework enables a structured comparison of different methods by linking method-specific evaluation metrics to interpretable aspects of multi-agent behavior.

Our analysis highlights several fundamental challenges inherent to offline multi-agent learning. First, limited state-action coverage and distributional shift significantly affect the performance of learned policies, particularly in high-dimensional and partially observable environments. Second, simple imitation learning approaches such as Behavioral Cloning provide strong and stable baselines, but often struggle to generalize beyond the demonstrated data. Third, while offline reinforcement learning methods offer a potential improvement, their effectiveness depends heavily on the quality of the dataset and the availability of suitable reward formulations.

Furthermore, our findings suggest that predictive accuracy alone is insufficient for modeling realistic multi-agent behavior. In particular, capturing coordinated team dynamics requires modeling temporal dependencies, spatial structure, and interactions between agents. Incorporating partial observability and domain-specific constraints is therefore essential for generating plausible and interpretable trajectories, in line with recent work on decentralized and constraint-aware modeling (Fujii et al., 2020).

Overall, this work demonstrates that offline multi-agent learning is a promising yet challenging approach for modeling complex real-world systems such as team sports. While existing methods are capable of capturing certain aspects of individual and collective behavior, achieving robust generalization and physically plausible coordination remains an open problem. The proposed evaluation framework provides a foundation for systematically analyzing these challenges and comparing future approaches in a consistent manner.

Future work may explore improved representation learning for partially observable settings, more effective reward design strategies grounded in domain knowledge, and hybrid approaches that combine offline learning with limited online interaction. In addition, incorporating explicit constraints and richer contextual information may further enhance the realism and applicability of learned policies.

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